Sealing Solutions

DATWYLER

COMPOUND DATA SHEET
FM140/1 Red

General description
FM140/1 is a chlorobutyl compound with silicate filler. Unconventionally cured.

Identification as per Ph.Eur. 3.2.9., method A
A typical ATR-FTIR spectrum of a clean, cut surface of FM140/1 is enclosed on page 3.

Physical properties

Hardness *ISO 48-4 (1s indentation; measured after min.1h)*	**46 ± 5 °Shore A**
Density *ISO 2781*	**1.323 ± 0.025 g/cm^3**
Ash content *Ph. Eur. 2.4.16*	**45.0 ± 2.0 %**
Compression set *ISO 815-1, (typical value)*	**32 %**
Modulus 100 *Internal method. 100mm/min, (typical value)*	**1.3 N/mm^2**
Modulus 300 *Internal method. 100mm/min, (typical value)*	**2.0 N/mm^2**
Elongation @ Break *Internal method. 200mm/min, ISO 37 Dumb-bell 2, (typical value)*	**670 %**
Tensile Strength @ Break *Internal method. 200mm/min, ISO 37 Dumb-bell 2, (typical value)*	**6.6 N/mm^2**
Water Vapor Transmission Rate *ASTM F-1249, 38°C, 100%RH (typical value)*	**0.0762 [g-mm]/m^2.24h**
Oxygen Transmission Rate *ASTM D-3985, 38°C, 90%RH, 100% O_2 (typical value)*	**90.17 [cc-mm]/m^2.24h**

Compound ingredient declarations

Latex	FM140/1 is not made with natural rubber latex.
Nitrosamines	FM140/1 is not made with chemicals that are associated with the formation of nitrosamines as listed in USP <1664.1>.
BSE/TSE	For the raw materials of FM140/1, certification confirming that such products are either of vegetable origin or are manufactured in severe process conditions for inactivation of prions as described in the EMA/410/01-rev.3 and in the European Pharmacopoeia 5.2.8. “Minimizing the risk of transmitting Animal Spongiform Encephalopathy Agents” is available.
MBT	FM140/1 is not made with 2-mercapto-benzothiazole (MCBT, also named MBT), or any of its derivatives.

Further compound declarations can be found in file MD0035.

For detailed extractables information, please contact your Datwyler representative.

Chemical properties

Ph.Eur.3.2.9.
ISO 8871-1
USP <381> : FM140/1 meets the chemical requirements for Type I closures specified in Ph.Eur. 3.2.9, USP <381> and ISO 8871-1. Typical results are given in the table on page 2. A typical UV spectrum is enclosed on page 3.

Ph.Jap.7.03. : FM140/1 meets the chemical requirements specified in Ph.Jap. 7.03. Typical results are given in the table on page 2. A typical UV spectrum is enclosed on page 3.

Note : Chemical testing is performed on non-post-treated standard test plates or products, prepared using representative process conditions.

Elemental impurities
An overview of the elemental impurities is given on page 2.

Biocompatibility

USP <87><88>:
ISO 10993-5
Ph.Jap.7.03.
FM140/1 is non-cytotoxic and meets the requirements of the Elution Test as described in USP <87>,”Biological Reactivity Tests, in vitro“ and the ISO 10993-5 "in vitro cytotoxicity". A typical test certificate is enclosed on page 4.
ISO 10993-5 cytotoxicity testing is considered equivalent with Pharm. Jap. 7.03 cytotoxicity testing.

Note : “FMxxx” refers to the type of compound, the extension “/x” refers to the color of the said compound. Differently colored compounds might be used for testing throughout this document. It is generally accepted that the color is irrelevant for the properties discussed, except Ash content and Density which are different per color.

Written by Hardeman, Tine, R&D, electronically signed
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Approved by Sharif, Tabassam, Regulatory Affairs, electronically signed

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Typical results of chemical properties as per Ph.Eur.3.2.9., USP <381> and ISO 8871-1 for FM140/1:

Characteristic		Limits	FM140/1
Appearance of solution	Turbidity	Type I: ≤ 6 NTU (*) Type II: ≤ 18 NTU (*)	**0.3**
	Color	Solution S is not more intensely colored than reference	**Pass**
Acidity or alkalinity (NA : Not applicable)		≤ 0.8 ml 0.01M HCl	**NA**
		≤ 0.3 ml 0.01M NaOH	**0.0**
UV Absorbance (max 220-360 nm)		Type I: ≤ 0.2 Type II: ≤ 4.0	**0.0**
Reducing substances		Type I: ≤ 3.0 ml 0.01M $Na_2S_2O_3$ Type II: ≤ 7.0 ml 0.01M $Na_2S_2O_3$	**0.2**
Extractable zinc (only for Ph. Eur.)		≤ 5.0 ppm Zn^{2+}	**0.4**
Ammonium		≤ 2 ppm NH_4^+	**Pass**
Residue on evaporation (only for Ph. Eur.)		Type I: ≤ 2.0 mg Type II: ≤ 4.0 mg	**0.2**
Volatile sulfides		Any black stain on the paper is not more intense than that produced by a control solution	**Pass**

(*) By definition corresponding with reference suspensions II and III (for Ph.Eur.) or B and C (for USP) respectively

Typical results of chemical properties as per Ph.Jap.7.03:

Characteristic		Limits	FM140/1
Appearance	%T at 430nm	≥ 99.0%	**99.6**
	%T at 650nm	≥ 99.0%	**99.8**
pH (difference with blank)		-1.0 ≥ ≤ 1.0	**-0.1**
Zinc		≤ 1 ppm Zn^{2+}	**0.4**
Reducing substances		≤ 2.0 ml 0.002 M $KMnO_4$	**0.7**
Residue on evaporation		≤ 2.0 mg	**0.2**
UV absorbance (Max. Abs. 220-350nm)		≤ 0.20	**0.02**
Cadmium*		≤ 5 ppm	**0.0**
Lead*		≤ 5 ppm	**1.1**

(*) measured directly on the rubber

Elemental impurities for FM140/1:

Elemental Impurity		ICH Q3D Class Reference	Result (µg/g)*
Arsenic	As	1	**<0.05**
Cadmium	Cd	1	**<0.05**
Mercury	Hg	1	**<0.05**
Lead	Pb	1	**<0.05**
Vanadium	V	2A	**<0.05**
Cobalt	Co	2A	**<0.05**
Nickel	Ni	2A	**<0.05**
Lithium	Li	3	**<0.05**
Copper	Cu	3	**<0.05**
Antimony	Sb	3	**<0.05**
Zinc	Zn	-	**70.47**

The elemental impurities are extracted with a 0.2N HNO_3, 0.05N HCl and 200 ppb gold solution in a 1g rubber/2.5mL ratio. The containers are sealed and heated to 70°C for 24h, after which they are allowed to cool and measured within 48h. The results are an average of 3 replicates, measured via ICP-OES according to Elemental Impurities – Procedures USP <233>.

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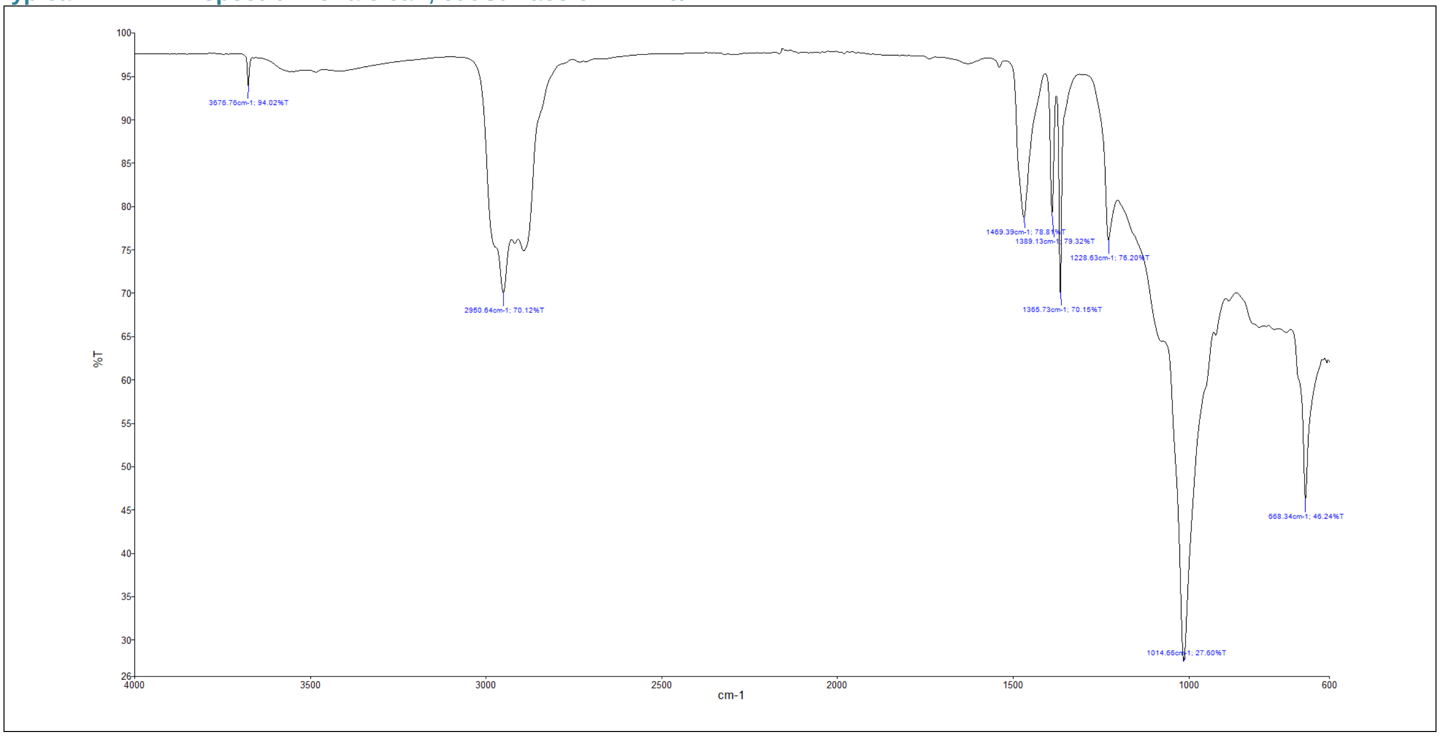
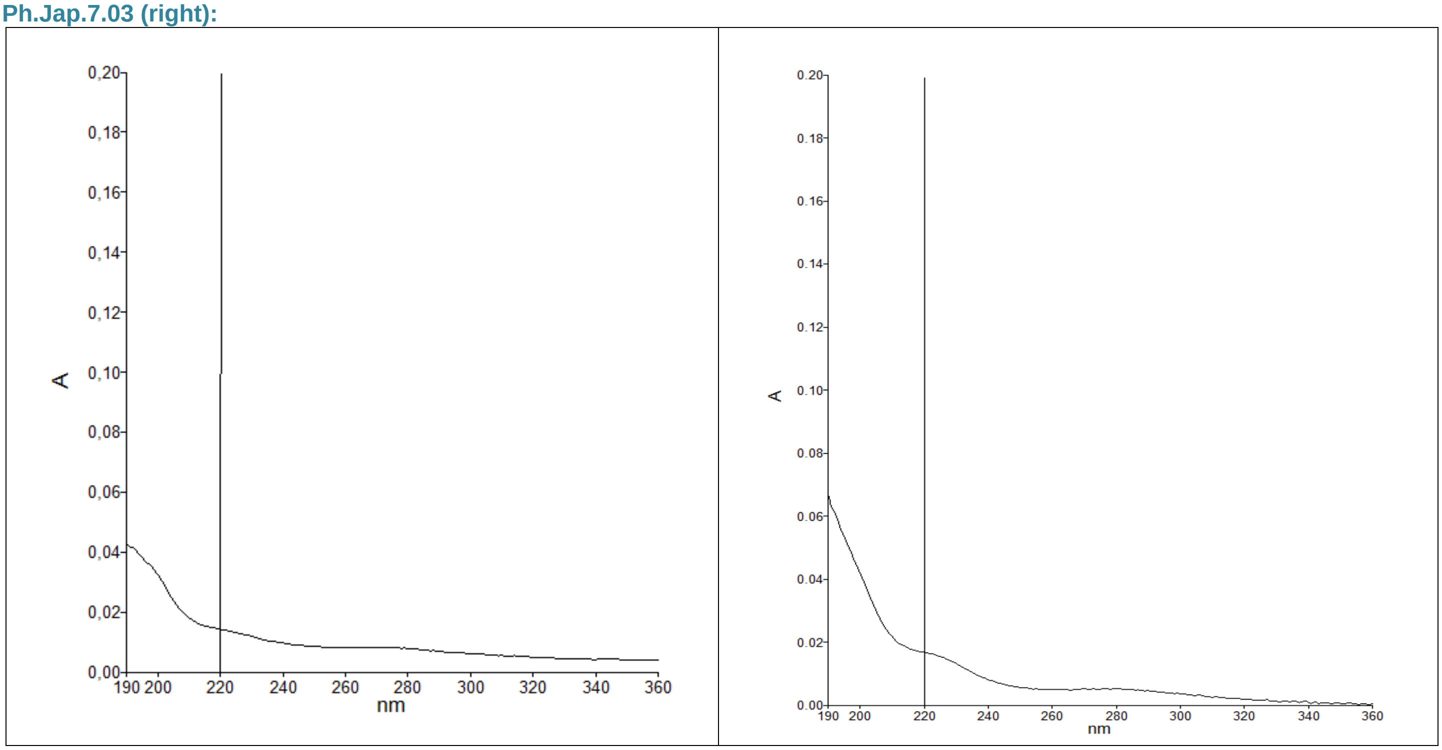
Typical ATR-FTIR spectrum of a clean, cut surface of FM140/1:



Typical UV spectrum of the Solution S extract of FM140/1, measured as per Ph.Eur. 3.2.9., USP <381> and ISO 8871-1 (left) and Ph.Jap.7.03 (right):



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USP<87> Elution Test certificate for FM140/1:

TOXIKON
Leaders in Life Science and Technology

TEST RESULT REPORT N°:10-B0277-N1

BELAC

Project Number:	**TE 10118**	**Study Number:**	**10-B0277-N1**
Sponsor:	**Helvoet Pharma Belgium NV**	**Report Date:**	**18/02/2010**
Contact:	*Mrs. Nadia Nouri*		
Address:	Industrieterrein Kolmen 1519	Date Sample Arrival:	10/02/2010
	3570 Alken, Belgium	Technical Initiation:	15/02/2010
PO.Number:	PB1000518	Technical Completion:	18/02/2010

Study	**Elution Test - ISO**	Temp/Time	37°C/24 hours
Test Item	FM140/0 V9025 SAF1	Ratio	25cm²/20mL
Lot	30194939	Vehicle	MEM-Complete

REFERENCE: According to "ISO 10993-5, 2009: Biological Evaluation of Medical Devices- Part 5:Tests for In Vitro Cytotoxicity." and "USP 32-NF 27, 2009: <87> Biological reactivity test, in vitro." Toxikon Reference: SOP 3.1.2.3, rev. 08

PROCEDURE: The biological reactivity of a mammalian monolayer, L929 mouse fibroblast cell culture, in response to the test item extract was determined. The samples and control articles were autoclaved prior to the preparation of the extracts. Extracts were prepared at 37±1°C for 24 hours in a humidified atmosphere containing 5±1% carbon dioxide (static). Positive (natural rubber) and negative (silicone) control articles were prepared to verify the proper functioning of the test system. The maintenance medium on the cell cultures is replaced by the extracts of the test item or control article in triplicate and the cultures are subsequently incubated for 48 hours, at 37±1°C, in a humidified atmosphere containing 5±1% carbon dioxide. Biological reactivity was rated on the following scale: Grade 0 (No reactivity); Grade 1 (Slight reactivity), Grade 2 (Mild reactivity), Grade 3 (Moderate reactivity) and Grade 4 (Severe reactivity). The test item is considered non-cytotoxic if none of the cultures exposed to the test item shows greater than mild reactivity (Grade 2).

OPINION AND INTERPRETATION: No reactivity (Grade 0) was exhibited by the cell cultures exposed to the test item at the 48 hours observation. Severe reactivity (Grade 4) was observed for the positive control article. The negative control article showed no signs of reactivity (Grade 0).

OPINION AND INTERPRETATION: Based on the evaluation criteria mentioned above, the test item is considered non-cytotoxic.

RECORD STORAGE: All raw data generated in this study will be archived at Toxikon Europe, according to SOP 4.2.8.

AUTHORIZED PERSONNEL

ir. Peter Cornelis
Study Director

Vanessa Ruymen
Quality Assurance

The test results on the enclosed report are only referring to the tested articles. Partly reproduction of this report can only be allowed after written permission of Toxikon. Toxikon guarantees that all results are acquired by testing according to officially accepted scientific methodology.

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